Customs Control Tower: A Practical Operating Model for Customs Excellence

📊 **Current dashboard:** Trade Data for Customs Excellence

Companion dashboard to this proposal: https://codebyann.github.io/customs-excellence-case-study/

📊 **Supporting dashboard:** Customs Control Tower / Tariff Exposure Analysis

Discussed with Mari during the 1st interview: https://codebyann.github.io/on-customs-control-tower/

Executive Summary

This proposal positions customs data as the foundation for a stronger, more scalable operating model. On already has the key document sources needed for better control: Purchase Orders, Commercial Invoices, Intercompany Invoices and Customs Declarations. The opportunity is to connect them at shipment and line-item level, compare critical fields, and create one trusted view of shipment quality, compliance risk, financial exposure and operational issues (one source of truth).

The dashboard is designed as a practical decision tool for Customs. It helps the team decide what needs further review, what needs escalation, what part can be automated, and where recurring issues should be fixed at the source (eg. repetitive flagged cases). This flow supports daily execution, audit readiness, stakeholder alignment and future automation.

It also fits On's broader data direction. Instead of building a separate customs reporting setup, customs data should be connected to the existing data environment, with BigQuery supporting trusted data and Looker supporting practical operating views.

Operating principle: start with trusted control logic, build the dashboard around action, and automate only where the process is ready.

1. Context and Business Problem

As On grows across markets, lanes, products, suppliers and shipment flows, customs operations become harder to manage. More movement creates more documents, more handovers, more classification decisions, more origin checks and more dependency on data quality.

The key risk is that different documents can describe the same shipment differently, often because of different naming conventions, timing or ownership across systems. A PO may show one quantity, the Commercial Invoice another value, the Intercompany Invoice another financial overview, and the Customs Declaration the official position submitted to authorities. If these views are not aligned, issues may only become visible later, for example during clearance, a broker query, financial reconciliation or an audit.

For a fast-growing company, this becomes an operating model question. Customs needs earlier visibility, clearer priorities and a repeatable way to decide which issues can be monitored, reviewed or escalated.

2. Proposed Approach

Most people do not process heavy data easily, not very well at that. We are not good at guessing odds or understanding instability. Teams need useful and user-friendly dashboards which help separate signal from noise and make the right decisions easier to see.

It starts with agreed control logic: which fields must match, which tolerances are acceptable, which differences create financial or compliance risk, and who owns the decision, resolution and evidence.

The next step is to connect the key customs data sources into one controlled shipment view:

- Purchase Order: commercial intent
- Commercial Invoice: external transaction basis
- Intercompany Invoice: internal financial flow
- Customs Declaration: official customs record
- Clearance outcome: operational result

And finally, create clear exception rules. A missing description field is not the same as an HS code mismatch, an origin inconsistency or a high-value duty variance. The process should separate low-risk data gaps from cases that may create delay, cost or audit exposure.

Behind the dashboard, I would define a **simple operating data model:** one shipment-level record, linked document tables, clear matching rules and an exception table. The exception table is critical because it creates the operational queue and the audit trail: what mismatch was found, who owns it, what decision was made and how it was resolved.

3. Dashboard Concept

The dashboard is designed to give Customs and related stakeholders one shared view of the shipment story. It connects document completeness, data quality, mismatch detection, risk prioritization, financial exposure and audit readiness.

The purpose is to define the right metrics and use them to help different teams make better decisions from the same trusted data source. Each main view should answer a practical question: Are we in control? Can we trust the data? What needs action now? Where is risk increasing? Where is cost exposure?

Decision moment	What the dashboard should help answer
Leadership	Control and trend visibility
Customs	What needs review and resolution first
Compliance	Risk patterns, evidence and audit readiness
Finance	Where mismatches create cost impact
Operations	Delay signals and root causes

The dashboard should also support audit-ready evidence at shipment level: documents, mismatch reason, rule triggered, owner, decision, resolution status and timestamp. This is relevant because strong customs control requires clear evidence of what was reviewed, why it was reviewed and what supported the final outcome.

4. Risk Scoring Logic

Risk scoring helps the team focus where expert action is required most. An effective customs process needs clear prioritization. Low-risk issues can be monitored, while cases with compliance, duty, clearance or audit impact should move into review or escalation.

The Customs Risk Score should combine five components: document mismatch, financial exposure, compliance sensitivity, clearance impact and repeated issue patterns. This helps the team focus on the issues with the highest business impact first. A shipment with high value, origin risk and clearance delay should be reviewed before a minor description issue with no financial impact.

The dashboard also supports a more effective operating model because it connects visibility with ownership. It should make clear what needs manual review, what needs escalation, what can be monitored, what can be automated and where root causes need to be fixed.

This is important because many customs issues do not originate inside Customs. They may come from master data, supplier inputs, invoice logic, broker handling, intercompany flows, product classification or unclear ownership between teams. Repeated patterns should inform supplier training, master data improvements, broker performance discussions and updated control rules.

5. AI, automation and future capabilities

AI should support customs experts, not replace their judgment. Customs decisions still require regulatory understanding, business context and accountability. AI is most useful when it reduces repetitive checking, highlights anomalies and guides users toward the right next step.

The most practical use cases are the ones closest to daily customs control: comparing documents, identifying value, quantity, HS code or origin differences, checking classification confidence, routing exceptions, highlighting likely root causes and preparing audit evidence. Automation should start with stable, repeatable checks such as document completeness, exact field matching, threshold-based value checks and owner routing.

A strong first automation project could focus on compliance document matching: comparing PO, CI, IC and CD fields, flagging mismatches, assigning ownership and keeping an audit-ready record of the outcome. The scope is focused enough to control, while still valuable enough to reduce manual review and prepare the process for future automation.

Phased roadmap:

• Phase 1 - Define control logic and key fields: critical fields, tolerances, ownership and risk triggers.

• Phase 2 - Build dashboard and exception views: shipment-level view, document completeness, mismatches and basic ownership.

• Phase 3 - Validate with Customs users: test whether exceptions are useful and prioritization supports real decisions.

• Phase 4 - Introduce risk scoring: prioritize by mismatch severity, financial exposure, compliance sensitivity, clearance impact and repeat patterns.

• Phase 5 - Automate selected checks: start with stable checks and keep human review for high-risk or unclear cases.

• Phase 6 - Scale and improve: use exception trends to improve master data, supplier instructions, broker performance, training and control rules.

6. Expected value and closing summary

The expected value is both operational and strategic. Operationally, the dashboard gives Customs earlier visibility, clearer priorities and a more consistent way to manage exceptions, helping reduce manual effort, clearance delays and avoid additional costs. Compliance benefits from stronger evidence and decision traceability. Finance gains better visibility into customs-related cost impact. The operating model becomes more ready for automation because checks are governed, owned and repeatable.

The key benefits are better visibility, earlier issue detection, stronger compliance control, reduced manual review effort, better prioritization of expert time, lower financial exposure, clearer accountability, stronger audit evidence and a better foundation for automation and AI support.

The core idea is to build a stronger customs operating workflow based on trusted data, clear control logic and practical decision support. Customs excellence means making customs data reliable enough to control risk, explain decisions and scale operations without adding manual work at the same pace as business growth.

Thank you for your time and for reviewing the proposal.

Sincerely,

Anna Kudyba